

COSC2446 Web Programming

AD006P24 - Associate Degree in Information Technology

COSC2446 Web Programming A

Assessment Number: 3 Extended Project & Demonstration			
Weighting	35%	Due Date	29 th of May 2026, COB (5:30 pm)
Course coordinator	Tanya Unterberger	Contact details	tanya.unterberger@rmit.edu.au (for general enquires only)
Individual or Group Assessment	Individual		

This assessment task is expected to take you a minimum of 40-50 hours to complete.

Task

This assessment includes a practical submission and a scheduled one-on-one demonstration.

Students submit a complete, working solution (full-stack website) based on the requirements outlined in the task. They are permitted to use AI tools to assist with development, provided all AI-generated content is appropriately acknowledged.

In the demonstration, you will walk through your submitted site, explain how key parts of your code work, and answer a few short questions related to the technologies used. You are allowed to use AI tools in your work, but you are expected to understand and be able to explain any code submitted.

Understanding Your Work

During your scheduled demonstration, you will be required to explain how key parts of your submitted code work. This is a vital component of the assessment.

If you are unable to demonstrate a clear understanding of your own code — regardless of whether it was written by you or with the assistance of AI tools — you will not be eligible to receive more than a **Pass (50%)** for the task.

This policy supports academic integrity and ensures that all students meet the learning outcomes of the course.

What is being assessed

In this assessment, the following course outcomes are being assessed:

1. Define the protocols and systems used on the Web
2. Explain the functions of clients and servers on the Web.
3. Create standards-compliant web pages.

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4. Implement client-side and server-side scripting.
5. Design and implement an interactive website with regard to issues of usability and accessibility.
6. Demonstrate the ability to access, modify, and present data using database persistence.

The following program outcomes are incorporated into this assessment task:

1. Enabling Knowledge
2. Critical Analysis
3. Problem Solving
4. Communication
5. Responsibility

Assessment details

This Extended Project aims to develop further PetConnect, a pet adoption platform using PHP, MySQL, and Bootstrap 5. The application enables owners to register, authenticate, and manage pet listings with image uploads. Both authenticated users and anonymous visitors will interact with various dynamic features, including a gallery with modal previews, full-text search, filtering by category, and secure CRUD operations.

The submission must demonstrate secure server-side development, client-side validation, reusable components, and adherence to best practices in code structure and deployment. The project must be completed and submitted individually.

This assessment includes a practical submission and a scheduled one-on-one demonstration.

AI TOOLS AND THEIR USE ARE NOT RESTRICTED FOR THIS ASSESSMENT TASK

Presentation requirements

In the demonstration, you will walk through your submitted site, explain how key parts of your code work, and answer a few short questions related to the technologies used.

Each student attends a scheduled session where they are asked to:

- Demonstrate the functionality of their submission.
- Explain key parts of their own code to verify their understanding.
- Answer short questions related to fundamental concepts covered in the assessment (HTML semantics, CSS layout, JavaScript logic, PHP, database queries, etc.).

The demonstration is not only to show functionality but to verify authorship and comprehension, supporting academic integrity in a context where AI use is permitted.

Assessment criteria

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You will be assessed on the following criteria:

Website Structure and Design	Website responsiveness
Database Structure and Design	GitHub and Deployment
Technical Implementation	Documentation and Submission
Session management	Adherence to requirements
Secure coding	Usability and validity

Refer to the assessment rubric for more details on grading.

Assessment 3 Checklist

#	Area	Requirement	Completed (Y/N)
Stage 1 – PHP Structure & Includes			
1	Structure	All required files exist in the a3 directory	
2	Structure	Required folders exist (assets, includes, assets/images/pets)	
3	Includes	includes/db_connect.inc supports localhost and Jacob 5	
4	Includes	header.inc, nav.inc, and footer.inc included on all pages	
5	Metadata	All pages have unique and meaningful <title> tags	
6	Semantics	Semantic HTML used (header, nav, main, footer)	
7	Layout	Responsive layout implemented with Bootstrap 5	
8	Navigation	Navbar includes PetConnect logo linking to home	
9	Navigation	Navbar includes a link to the search page with a Material Icon	
10	Layout	Footer includes your name and gradient background	
Stage 2 – Database Structure & CRUD Operations			
11	Database	Database created on Jacob 5 using correct schema	
12	Database	Table and column names match provided schema exactly	
13	Security	All database queries use prepared statements	
14	Home Page	index.php shows carousel of latest images from DB	

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15	Home Page	<code>index.php</code> shows grid of latest 4 pets from DB
16	Pets Page	<code>pets.php</code> uses two-column layout with DB records
17	Pets Page	Pet names link to <code>details.php</code>
18	Details Page	<code>details.php</code> displays a single record via query string (prepared statements)
19	Gallery	<code>gallery.php</code> displays grid of images from database
20	Gallery	Each image links to corresponding details page
21	Search	<code>search.php</code> performs full-text search (prepared statements)
22	Owner page	<code>owner.php</code> displays pets by owner via query string (prepared statements)
Stage 3 – JavaScript, Modals & Client-Side Logic		
22	JavaScript	Single custom JS file used (<code>assets/js/scripts.js</code>)
23	JavaScript	No inline JavaScript used
24	JavaScript	Image extension validation implemented in JS
25	JavaScript	Only jpg, jpeg, png, gif, webp accepted
26	JavaScript	Clicking image opens Bootstrap modal – gallery and details pages
27	JavaScript	Client-side category filter implemented (JS only)
28	JavaScript	Confirmation modal opens from details page before record deletion
Stage 4 – Authentication & Sessions		
29	Authentication	<code>register.php</code> creates users securely (hashed passwords). Upon registration the user is automatically logged in.
30	Authentication	<code>login.php</code> authenticates users using sessions
31	Authentication	<code>logout.php</code> destroys session and redirects correctly
32	Sessions	Flash messages implemented using PHP sessions
33	Access	Authenticated vs anonymous views handled correctly
Stage 5 – Authorization & Protected Routes		

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34	Protection	<code>add.php</code> accessible only to logged-in users
35	Protection	<code>edit.php</code> accessible only to record owner
36	Protection	Delete action accessible only to record owner
37	Protection	Delete removes both DB record and image file
38	Ownership	Owner-only edit/delete buttons enforced server-side
39	Files	Uploaded images use server-generated unique filenames (e.g. <code>uniqid()</code> or timestamp-based naming)
Validation, Deployment & Security		
40	Validation	All HTML pages validate via W3C Validator
41	Validation	CSS validates via W3C CSS Validator
42	Security	<code>assets/images/pets/</code> added to <code>.gitignore</code>
43	Deployment	Site deployed to Coreteaching server
44	Deployment	Permissions set (755 on <code>a3</code> , 777 on images folder)
45	Deployment	All pages and assets load with no errors
46	Security	<code>.htaccess</code> protects against unauthorised viewing of your site
GitHub, Submission & Demo		
46	Git	Project located in <code>a3</code> directory of GitHub repo
47	Git	Git history shows consistent work over time
48	Git	Teacher added as a collaborator
49	Submission	ZIP named <code>COSC2446_a3_<studentID>.zip</code>
50	Submission	ZIP includes all source files and assets
51	Documentation	<code>README.md</code> includes live site URL
52	Demo	Able to explain PHP, DB queries, sessions, and security
Submission requirements		
You are required to submit your completed project as follows:		

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Deployment

- The live site must be deployed to a **Coreteaching web server** – Jupiter/Saturn/Titan (jupiter.csit.rmit.edu.au/saturn.csit.rmit.edu.au/titan.csit.rmit.edu.au).
- The database for the live deployment must be created on **Jacob 5**, named **exactly** after your student ID.
- Ensure all links, assets, and dynamic content function correctly on the deployed site.

Canvas Submission

Submit the following as a zipped folder named COSC2446_a3_<yourStudentID>.zip via **Canvas** under the “AT3: Extended Project & Demonstration” assignment:

- All source files and assets.
- A README.md file containing:
 - The **URL of the live site**.

Academic integrity

It is important that you submit material which is your own work. The University provides information for students on academic Integrity through the [RMIT website](#). If you are unsure about academic matters please be sure to ask your teacher for information.

Assessment Policy

Assessments submitted after the due date will be accepted as follows:

- Assessments submitted between 0-24 hours after the due date: 20% penalty
- Assessments submitted between 24-48 hours after the due date: 50% penalty
- Assessments submitted more than 48 hours after the due date will not be accepted

Extensions

If you will not be able to meet the due date for an assessment you may apply to your teacher for an extension of up to seven days by completing the Application of Time to Submit Assessment Work Form at

https://www.rmit.edu.au/content/dam/rmit/documents/Students/Student_forms/Application-for-extension-of-time-to-submit-work.pdf Applications for an extension of time must be received at least one working day before the due date for an assessment.

Special Consideration

If unforeseen circumstances beyond your control prevent you from submitting your work on time, you may be eligible to apply for special consideration. For further information regarding special consideration, please refer to the RMIT Special Consideration page at <https://www.rmit.edu.au/students/student-essentials/assessment-and-results/special-consideration>

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Functional Requirements

Code and Git Usage

- All code should reside in a “wp” GitHub repository that is under your RMIT student ID username
- The original clone is from <https://github.com/tanyarmit/wp.git>. Add your teacher as a collaborator.
- Work in the a3 directory.
- Use meaningful commit messages and push frequently.
- README.md file should contain the link to your Coreteaching deployment.
- Refer to “Developing and deploying with Git” for instructions on uploading and testing files on Coreteaching web servers.
- Make sure that you follow the naming convention for files and directories. Failure to do so will result in lost marks.
- Students must demonstrate consistent use of Git throughout the project timeline. A minimum of 5 commits spread over at least 4 separate days is expected. No single day can contain more than 50% of the work.

Penalty:

- If there are fewer than 5 commits, or all or 50% of the commits are made on the same day, a 50% penalty will be applied to the overall marks.
- This is to encourage regular progress, documentation of development steps, and authentic authorship.

Page structure & Design

- Match the provided screenshots exactly
- Use semantic HTML5: <header>, <nav>, <main>, <footer>
- Consistent navbar with links, search, PetConnect that is a link to the Home page
- Use Google Material Icons for the search icon.
- Footer includes a copyright with your name
- Responsive design using Bootstrap 5 for all pages
- Validate all HTML and CSS - <https://validator.w3.org/> and <https://jigsaw.w3.org/css-validator/>.

Database & Connection

- To create a petconnect database on localhost, use the SQL schema provided on Canvas. For Jacob 5, the database name should be your student ID. Using different tables or column names will lead to a fail mark for the assignment.
- All database access should be performed via **MySQLi procedural** and **prepared statements**.
- Single connection in includes/db_connect.inc, with error reporting for development. The file should cater for access to either localhost (when run via XAMPP) or Jacob 5 - talsprddb02.int.its.rmit.edu.au

Common Includes & Assets

- **includes/ folder:**
 - header.inc: Bootstrap & custom CSS, Material Icons, favicon links, flash-message display

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- `nav.inc`: navigation links, search form with Material Icon button
- `footer.inc`: gradient background, copyright with your name
- `db_connect.inc`: handles database connections to localhost and Jacob5
- Helpers (optional): `functions.inc`, etc.
- **Fonts**: Use Google fonts.
 - Headings*: Poppins - <https://fonts.google.com/specimen/Poppins>
 - Weights: 400, 500, 600, 700, 800
 - Usage: All heading elements (h1-h6), navbar brand
 - Body Text*: Inter - <https://fonts.google.com/specimen/Inter>
 - Weights: 400, 500, 600, 700
 - Usage: Body copy, form labels, buttons, navigation links
 - Icons*: Material Icons - <https://fonts.google.com/icons>
- **CSS**: one common custom CSS file for all pages - `assets/css/styles.css`.
- **JS**: one common custom JS file for all pages - `assets/js/scripts.js`. Used for file extension validation, record filtering, deletion confirmation and modals.
- **Images**: Images like the banner and favicon should reside in `assets/images`. Pet-related images uploaded by users should reside in `assets/images/pets`.
- **Colours**: The following colours are used for the site:
 - Brand Colors*
 - Primary Color: #6366f1 (Indigo)
 - Primary Dark: #4f46e5 (Deep Indigo)
 - Primary Light: #818cf8 (Light Indigo)
 - Secondary Color: #ec4899 (Pink)
 - Secondary Dark: #db2777 (Deep Pink)
 - Accent Colors*
 - Accent Orange: #f59e0b
 - Accent Green: #10b981
 - Accent Purple: #8b5cf6
 - Neutral Colors*
 - Text Dark: #1f2937 (Dark Gray)
 - Text Light: #6b7280 (Medium Gray)
 - Background Light: #f9fafb (Off White)

File structure in `a3` folder

```

├─ assets/                # Static assets (images, CSS, JS)
|   └─ images/
|       └─ pets/          # Pet photos
|   └─ css/
|       └─ styles.css     # CSS
|   └─ js/
|       └─ scripts.js     # JavaScript

```

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└─ includes/	# Include files (header, footer, nav, db connection)
└─ db_connect.inc	
└─ header.inc	
└─ footer.inc	
└─ nav.inc	
└─ index.php	# Homepage with carousel and pet listings
└─ details.php	# Pet details page
└─ gallery.php	# Gallery view of all pets
└─ pets.php	# Browse pets by species
└─ register.php	# User registration
└─ login.php	# User login
└─ logout.php	# User logout
└─ owner.php	# Displays all pets by the same owner
└─ add.php	# Add new pet form. Requires login
└─ process_add.php	# Add pet handler. Not mandatory. Adding a new record can # be handled by add.php
└─ edit.php	# Editing pet's record. Accessible by the logged in owner
└─ search.php	# Shows search results

Pages & Features

File	Requirements
index.php	- Bootstrap carousel of the 4 latest images - Responsive grid of 4 latest pets
register.php + processing	- Form & script using <code>password_hash()</code> encryption - Flash feedback via <code>\$_SESSION</code> - Immediately log in user upon successful registration
login.php + processing	- Form & script using <code>password_hash()</code> encryption - Flash feedback via <code>\$_SESSION</code>
logout.php	- Destroys session → redirects to the home page
add.php + processing	- Protected (login only) - Form & adding a record script. Use prepared statements - Server-side checks, record creation and image upload

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	Uploaded image files must use unique filenames to prevent overwriting existing files. Acceptable approaches include using the PHP <code>uniqid()</code> function or prefixing or suffixing the original filename with a timestamp (e.g., <code>time()</code> or <code>date()</code> functions)
edit.php + processing	- Protected & owner-only - Replaces old image file on upload - Cleans up old file
pets.php	- Two-column layout: left - banner image, right - table - Table with records from the database - Custom link colours - Name is a hyperlink to <code>details.php</code>, which displays selected record
gallery.php	- Grid of all pet images - Names of the images come from the database. The actual files are in <code>assets/images/pets</code> - Client-side filter by category (JavaScript only) - Images open in a Bootstrap modal on click
details.php	- Detail page with pets image. Use prepared statements - Edit/Delete buttons for owner only. Protected. - Deletion requires confirmation, removes associated image file and the DB record
owner.php	- Shows all their pets via cards. Use prepared statements - No login required to view the page - A link to the owner with a query string is displayed on the <code>details.php</code> page
search.php	- Full-text search on title/description - Prepared statements

Client-Side JavaScript

- **Image-extension validation** on for the `<input type="file" name="image">` field: only jpg, jpeg, png, gif, webp.
- **Image preview** before the selected file is uploaded on the **add.php** page.
- **Modals** for full-size images **gallery.php** (no inline JS—use `assets/js/scripts.js`).
- **Category filter** on **gallery.php**: pure JavaScript show/hide. Use `data-status` field.
- **Deletion confirmation** on **details.php** when owner click the “Delete” button.

Security & Best Practices

- Prepared statements everywhere. All scripts that use queries with user input, i.e. inserting records (including registration), searching for records (including detail pages, search, login, etc.), editing and deleting records must be secure, and use SQL injection prevention methods – prepared statements.
- Flash messages must use PHP sessions for all user interactions (`success/error`).
- File-system hygiene: new uploads are created, old images are deleted on edit/delete.

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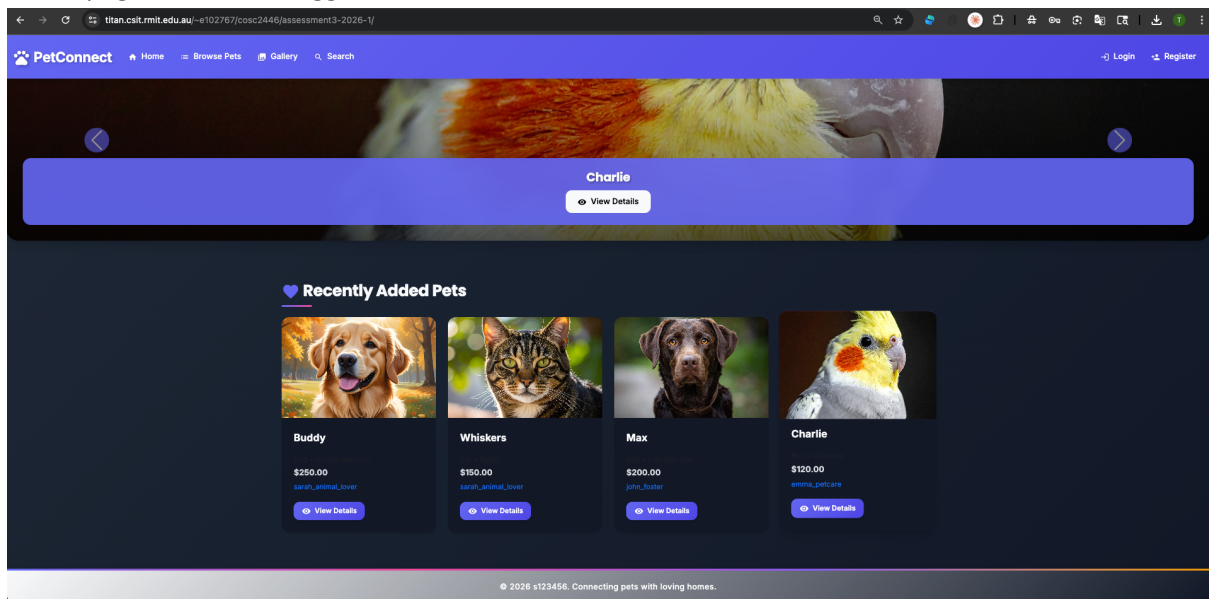
- Maintain clean uploads (e.g. unique file names). You must not rely on the original uploaded filename. A unique name must be generated server-side using methods such as `uniqid()` or a timestamp-based naming convention.
- Add `assets/images/pets/` to `.gitignore` to prevent sync issues.

Submission Instructions

1. Push to your private Git repository with clear commits.
2. Deploy to the Coreteaching web servers and ensure all assets load correctly.
3. Use the `.htaccess` file to prevent unauthorised access to your site on Coreteaching servers.
4. Submit via Canvas a zipped folder with:
 - All files
 - Live site URL in README.md

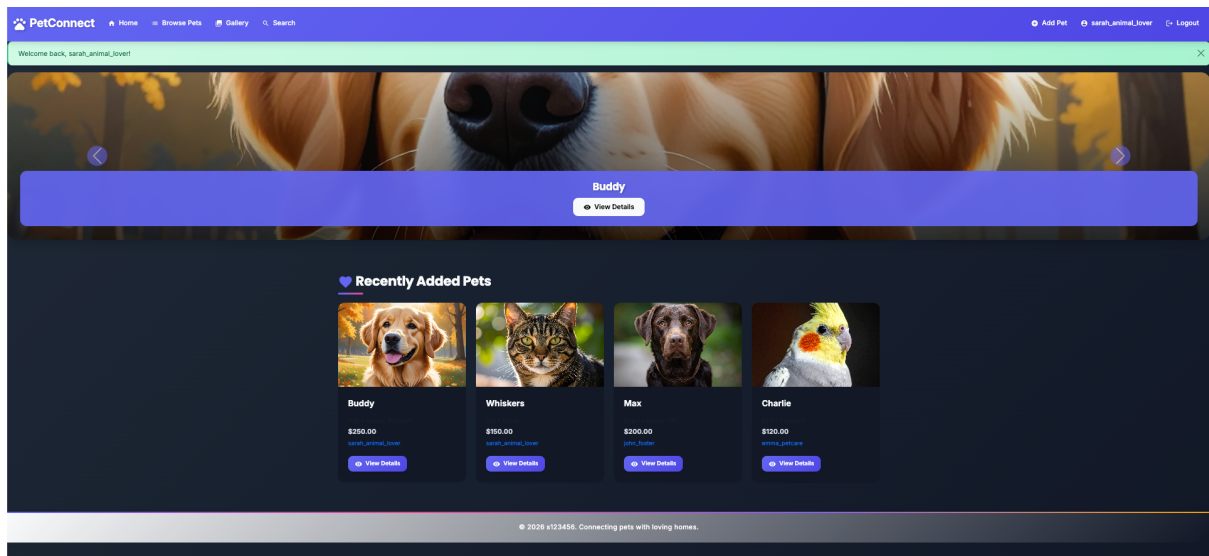
NOTE: The following screenshots are provided as reference for the required look and feel. Students are expected to replicate these exactly in their implementations.

Home page, user is not logged in:

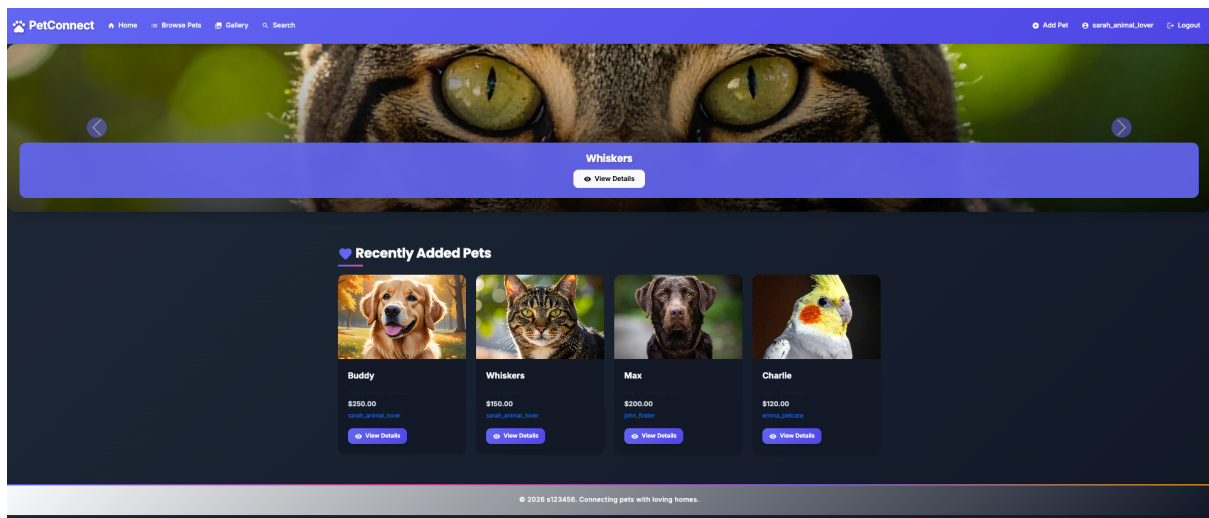


Home page, the user has just logged in:

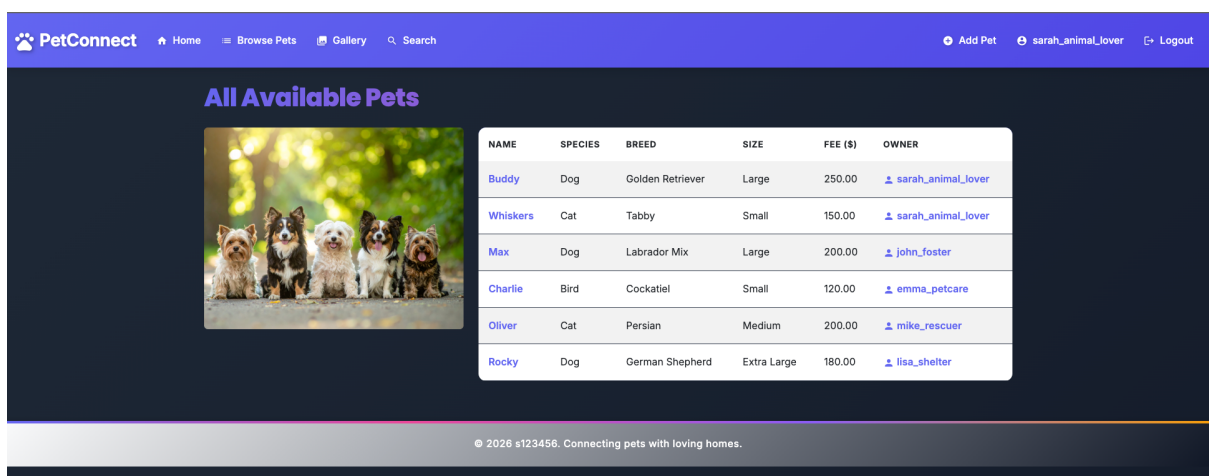
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Home page, user is logged in:

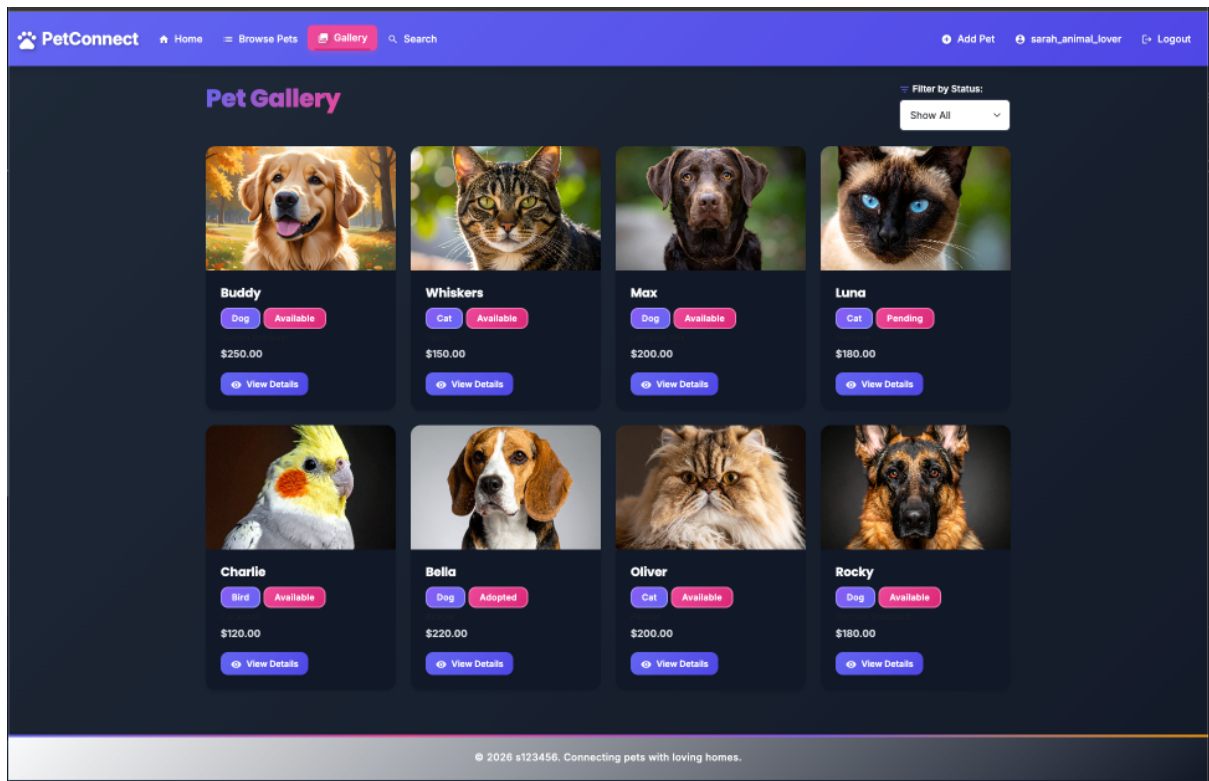


Pets page:

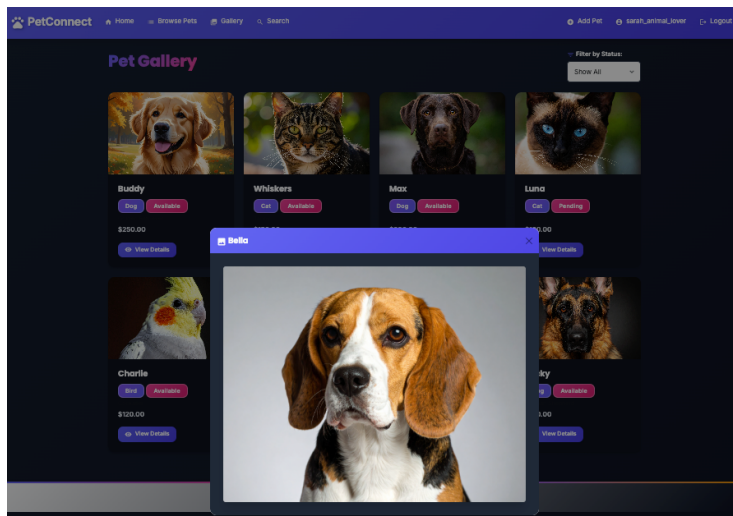


Gallery page:

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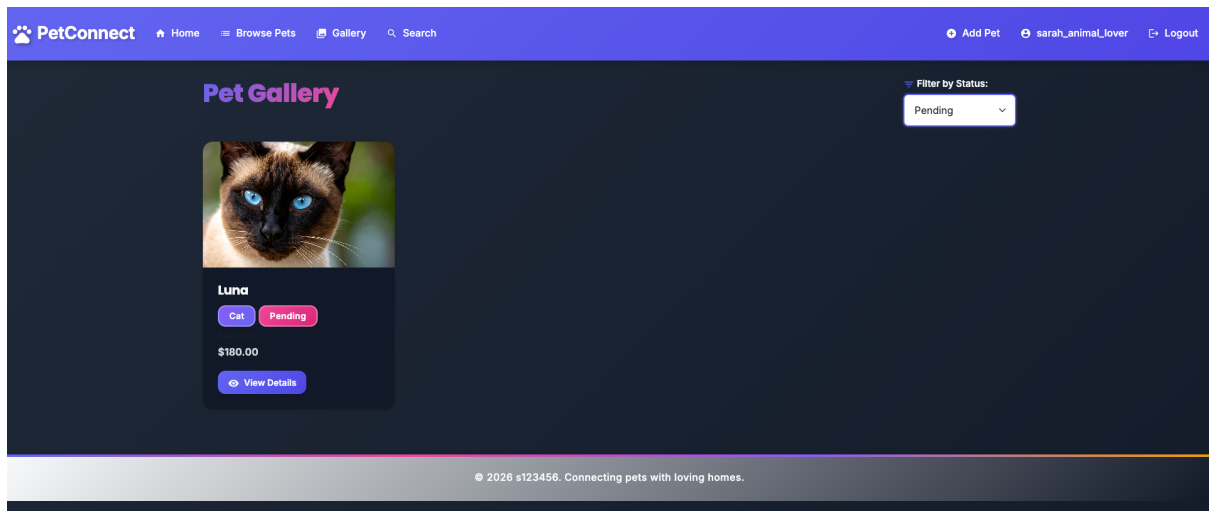


Gallery page, with modal:

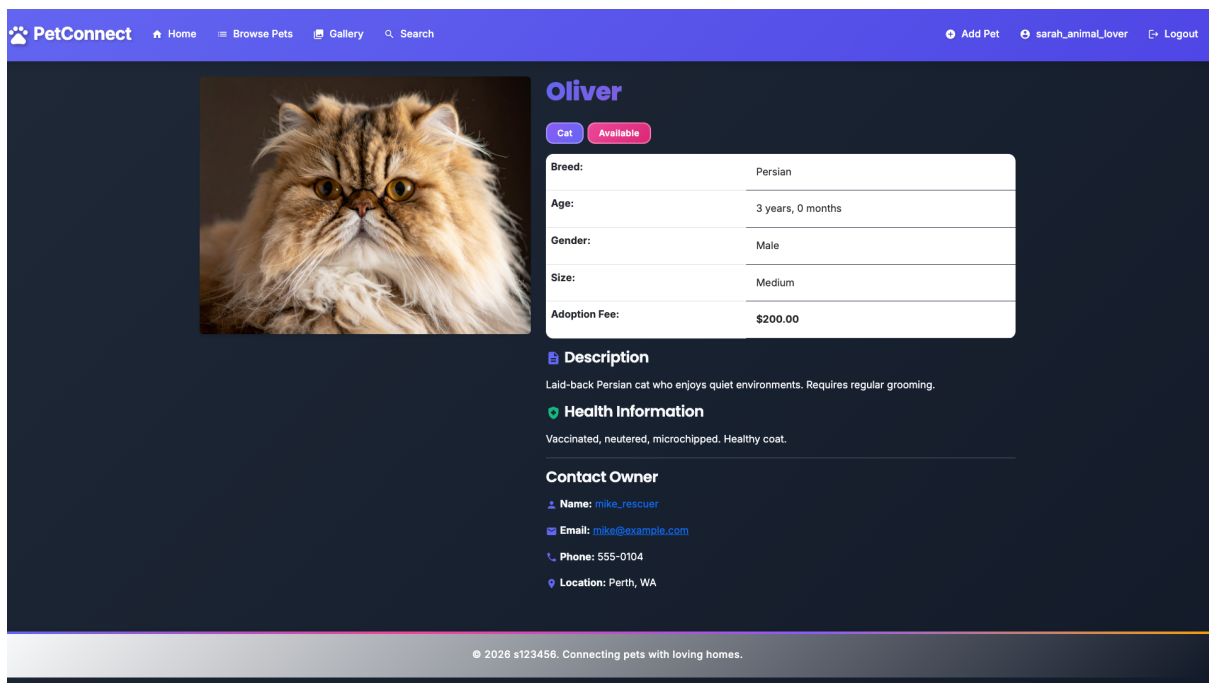


Gallery page with a filter applied:

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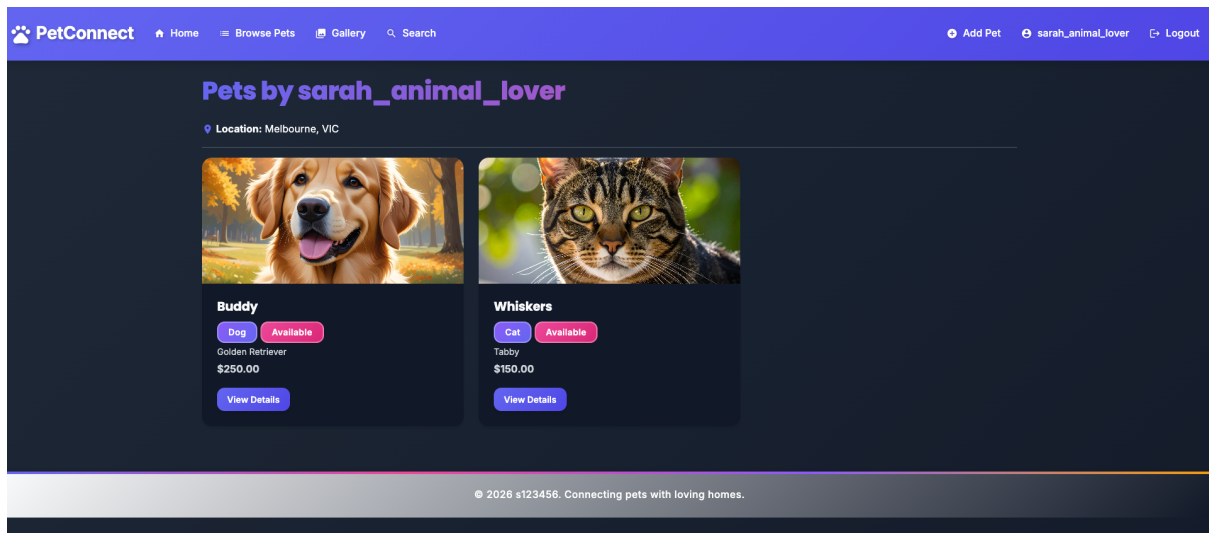


Details page, user is not logged in or not the owner of the record:



Owner page:

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Add page, the user is logged in:

The screenshot shows the "Add a New Pet for Adoption" form on the PetConnect website. The form includes the following fields and controls:

- Pet Name ***: Text input field.
- Species ***: Dropdown menu with "Select species" as the placeholder.
- Breed**: Text input field.
- Age (Years)**: Text input field.
- Age (Months)**: Text input field.
- Gender ***: Dropdown menu with "Select gender" as the placeholder.
- Size ***: Dropdown menu with "Select size" as the placeholder.
- Adoption Fee (\$) ***: Text input field.
- Description ***: Large text area for the pet's description.
- Health Information**: Text area for health-related details.
- Status ***: Dropdown menu with "Select status" as the placeholder.
- Pet Photo**: File upload section with a "Choose file" button.
- Buttons**: "Add Pet" (blue) and "Cancel" (red) buttons at the bottom.

The footer contains the copyright notice: © 2026 s123456. Connecting pets with loving homes.

Details page, the user is logged in and the owner of the record:

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 [Home](#) [Browse Pets](#) [Gallery](#) [Search](#) [Add Pet](#) [sarah_animal_lover](#) [Logout](#)



Buddy

Dog Available

Breed:	Golden Retriever
Age:	3 years, 6 months
Gender:	Male
Size:	Large
Adoption Fee:	\$250.00

Description

Friendly and energetic golden retriever looking for an active family. Loves playing fetch and swimming.

Health Information

Fully vaccinated, microchipped, neutered. No known health issues.

Contact Owner

Name: sarah_animal_lover
Email: sarah@example.com
Phone: 555-0101
Location: Melbourne, VIC

[Edit](#) [Delete](#)

© 2026 s123456. Connecting pets with loving homes.

Edit page, the user is logged in and the owner of the record:

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PetConnect

Home

Browse Pets

Gallery

Search

Add Pet

sarah_animal_lover

Logout

Edit Pet: Whiskers

Pet Name *

Whiskers

Species *

Cat

Breed

Tabby

Age (Years)

2

Age (Months)

0

Gender *

Female

Size *

Small

Adoption Fee (\$) *

150.00

Description *

Sweet and affectionate tabby cat. Great with children and other cats. Indoor cat preferred.

Health Information

Vaccinated, spayed, microchipped. Healthy.

Status *

Available

Current Photo



Update Pet Photo (Optional)

Choose file

Update Pet

Cancel

© 2026 s123456. Connecting pets with loving homes.

Details page, the user is logged in and the owner of the record. Delete button clicked:

PetConnect

Home

Browse Pets

Gallery

Search

Add Pet

sarah_animal_lover

Logout



Whiskers

Cat

Available

Breed:	Tabby
Age:	2 years, 0 months
Gender:	Female
Size:	Small
Adoption Fee:	\$150.00

Description

Sweet and affectionate tabby cat. Great with children and other cats. Indoor cat preferred.

Health Information

Vaccinated, spayed, microchipped. Healthy.

Contact Owner

Confirm Deletion

Are you sure you want to delete Whiskers's listing?

This action cannot be undone.

Cancel

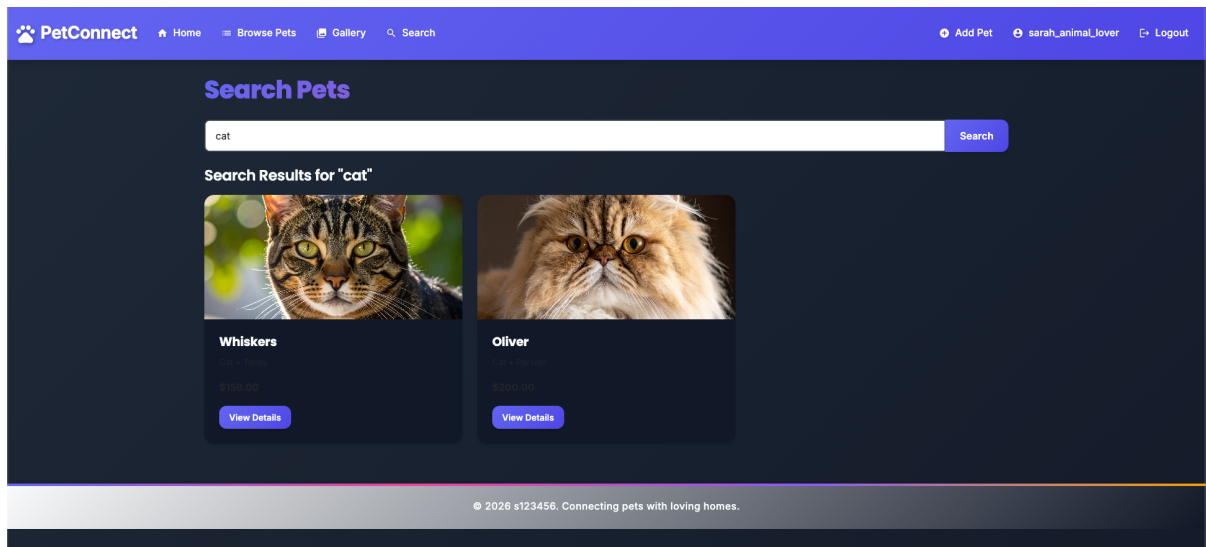
Yes, Delete

© 2025 PetConnect. Connecting pets with loving homes.

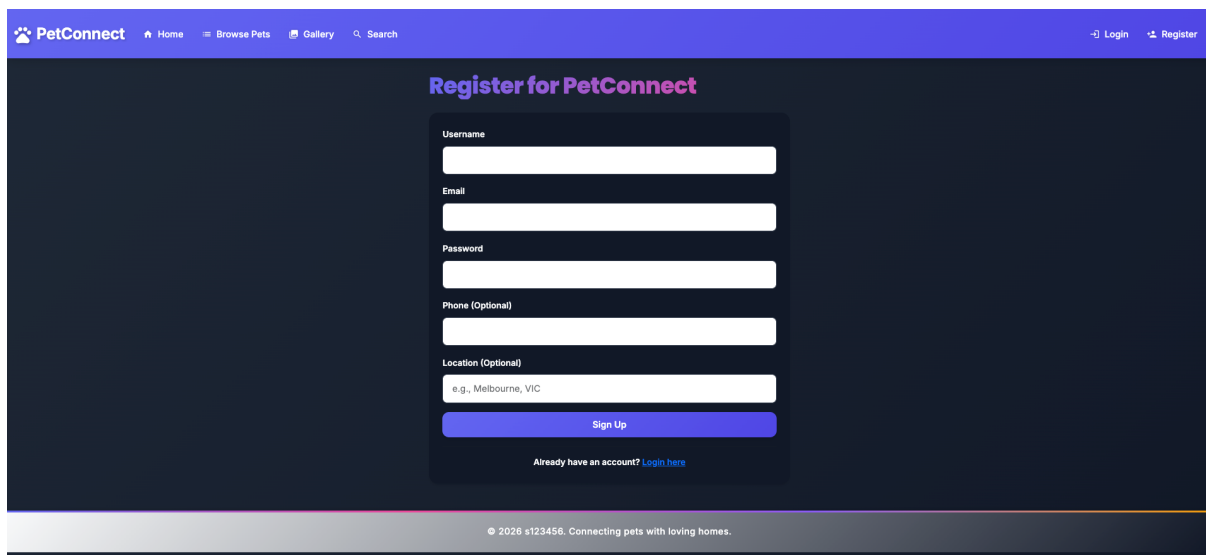
Search page:

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Register page:



Login page:

